

Shiyao Tang

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Education

M.S., Transportation Engineering

University of California, Los Angeles; GPA: 4.0/4.0

Los Angeles, CA

Sept. 2025 – Present

B.Eng., Traffic and Transportation

Beijing Jiaotong University (BJTU); WES GPA: 3.83/4.0

Beijing, China

Sept. 2021 – June 2025

Exchange Student, Transportation and Operations

University of California, Irvine; GPA: 4.0/4.0

Irvine, CA

Mar. 2024 – Aug. 2024

Exchange Student, Industrial and Systems Engineering

The Hong Kong Polytechnic University; GPA: 4.0/4.0

Hong Kong

Sept. 2024 – Jan. 2025

Research Interests

- **Topics:** Transportation demand modeling, travel behavior, network optimization, smart mobility systems
- **Methods:** Discrete choice modeling, convex optimization, graph-based learning, reinforcement learning

Research Manuscripts

- **Tang, S., & Zhou, X.** “Adaptive Test-Time Learning for Driver Fatigue Detection.” *IEEE Transactions on Intelligent Transportation Systems*. **Major revision**.
- Meena, K. K., **Tang, S.,** & Kim, Y. “PD-MUSE: Primal–Dual Maximum-Utility Structured Estimator for Discrete Choice Models.” Manuscript in preparation for *Transportation Research Part B: Methodological*.
- **Tang, S.,** Ji, K. Y., Bills, T., & Kim, Y. “PD-MUSE: Primal–Dual Maximum-Utility Structured Estimator with Entropy Regularization and Moment Constraints.” Manuscript in preparation for *Transportation Research Part A: Policy and Practice*.

Peer-Reviewed Publications

- **Tang, S.** “Review on Short-Term Traffic Flow Prediction Methods under Big Data.” *International Conference on Mechatronics and Smart Systems*, 2023.

Conference Presentations

- **Tang, S.,** Ji, K. Y., Bills, T., & Kim, Y. “PD-MUSE: Primal–Dual Maximum-Utility Structured Estimator with Entropy Regularization and Moment Constraints.” **Accepted for presentation**, 2026 *INFORMS Annual Meeting*, San Francisco, CA, Nov. 1–4, 2026.
- **Tang, S.** “Convex Reformulations for Logit-Family Choice Models: A Primal–Dual Estimator and Open-Source Implementation.” Talk, *The Fourth ITS Irvine Symposium on Emerging Research in Transportation*, University of California, Irvine, 2026.

Research Experience

Graduate Research Assistant

University of California Institute of Transportation Studies (UC ITS)

Los Angeles, CA

July 2026 – Present

Advisor: Youngseo Kim

- Collaborate with Kapil Meena and Youngseo Kim on bi-level and one-shot PD-MUSE formulations for MNL, NL, and IPDL, including MNL maximum-likelihood equivalence and structured-model score decomposition.
- Develop empirical applications of PD-MUSE using UCLA Mobility stated- and revealed-preference data.
- Explored maximum-entropy route-choice modeling using the LA-Sim network and PeMS traffic detector data.

Undergraduate Thesis

“Mapping Trip Distance Distributions Using Network Embedding and Topological Analysis”

Supervised by Irene Martinez (TU Delft) and Yonglei Jiang (BJTU)

- Developed a cross-city framework to analyze and predict trip distance distributions using OpenStreetMap-based network features across cities in China and the Netherlands.
- Combined network embedding, topological analysis, and gravity-style modeling to study how transportation network structure shapes urban travel behavior.

Core Member, Adaptive Test-Time Learning (ATT)

Nanyang Technological University

Project: Driver Fatigue Detection; supervised by Xinliang Zhou and Yang Liu

- Developed subject-specific pattern assessment and adaptive learning rates for cross-driver fatigue detection on NTHU-DDD, UTA-RLDDD, and FAHD, improving generalization under distribution shift.
- Built a WandB, t-SNE, and Matplotlib pipeline to track adaptation and cross-subject fatigue-pattern alignment.

Research Assistant

Freight Train Identification Project with Caltrans

Supervised by Stephen G. Ritchie

- Extended more than 1,000 night-time recognition labels and prepared freight-train images for model refinement.
- Improved night-time label recognition through deep learning model tuning, GIS workflows, and signal processing.

Research Member

Cambridge University Academic Training Program

- Built robot navigation and emergency-delivery simulations in Webots, contributing to path planning, behavioral decision-making, algorithm design, and deployment.
- Developed a heterogeneous robot-team framework that reduced simulated urban delivery time by 25% and improved task completion efficiency by 40%.

Industry Experience

Cloud Control Technology Department Intern

Western China Internet of Vehicles (Chongqing) Co., Ltd.

- Supported the Chongqing Intelligent Networked New Energy Vehicle Demonstration Zone Project through site surveys, IoV equipment commissioning, and technical coordination.
- Assisted in drafting the *Vehicle, Road, and Cloud Integration System White Paper 3.0*.

Vehicle-Road Coordination R&D Intern

China Merchants Chongqing Communications Technology Research and Design Institute

- Compiled international standards and case studies for intelligent transportation and smart city applications.
- Authored project proposals and field survey reports; presented findings to senior management.

Selected Awards & Honors

- **Best Diplomatic Presence Award**, 2023 “Model APEC” Conference
- **Beijing Municipal Second Prize**, “Internet+” Innovation and Entrepreneurship Competition
- **First Prize**, BJTU Innovation & Entrepreneurship Qualification Trials
- **School Class Scholarship for Academic Excellence**, First Prize (top 1%)

Additional Information

- **Languages:** Mandarin (native), English (IELTS: 8; GRE: 335)
- **Technical Skills:** Python, R, Julia, MATLAB; Vissim, MATSim, TransModeler, Webots, Civil3D, AutoCAD
- **Interests:** Public speaking, tennis, volleyball, guzheng, guqin, calligraphy